**RADIX SORT VISUALIZER**
**APPLICATION DOCUMENTATION (USER MANUAL)**

# 1. INTRODUCTION

The Radix Sort Visualizer is a desktop application developed using Python and Tkinter. The program is designed to visually demonstrate how the Radix Sort algorithm works by displaying each step of the sorting process.

Users can manually enter numbers or generate random values and observe how the algorithm sorts the data through animated visualizations. The application provides controls that allow users to start, stop, resume, move through previous and next steps, and clear the visualization.

This manual provides complete instructions for installing, operating, and understanding the features of the application.

# 2. PURPOSE OF THE APPLICATION

The Radix Sort Visualizer was developed to help users understand the Radix Sort algorithm through interactive visualization.

The application allows users to:

- Enter numerical values to be sorted
- Generate random numbers automatically
- Observe each stage of the sorting process
- Navigate through sorting steps manually
- Pause and resume the animation
- Review the final sorted result

# 3. SYSTEM REQUIREMENTS

Before running the application, ensure that the following requirements are met.

**Hardware Requirements**

- Computer or laptop
- Minimum 4 GB RAM
- Minimum 100 MB available storage

**Software Requirements**

- Python 3.10 or newer
- Tkinter library
- Windows, Linux, or macOS operating system

# 4. STARTING THE APPLICATION

**Step 1: Open the Project Folder**

Locate the folder containing the following files:

- main.py
- gui.py
- radix.py
- utils.py

**Step 2: Run the Program**

Open a terminal or command prompt.

Navigate to the project folder.

Execute the following command:

python main.py

**Step 3: Application Launch**

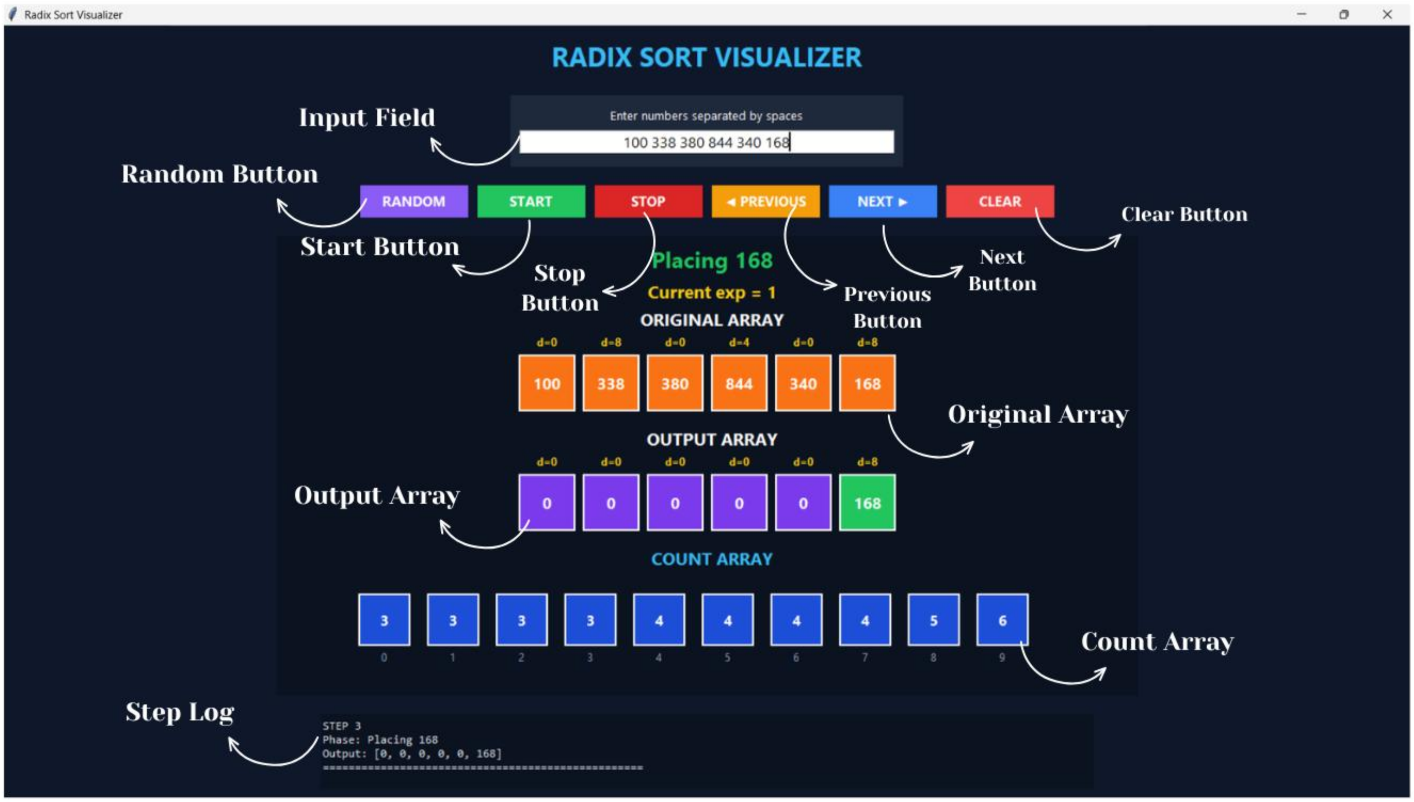
After execution, the Radix Sort Visualizer window will appear.

The main interface consists of:

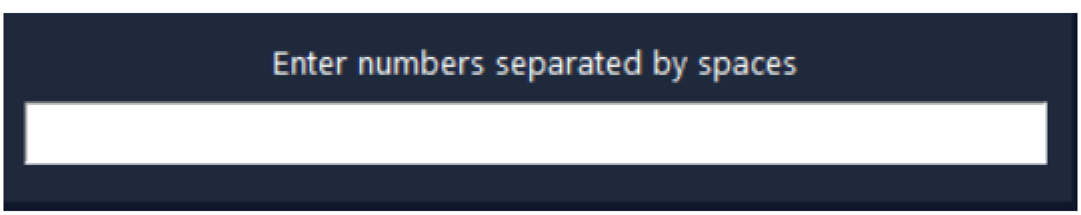
- Input Field
- Random Button
- Start Button
- Stop Button
- Previous Button
- Next Button
- Clear Button
- Visualization Area

- Output Log Panel

# 5. USER INTERFACE GUIDE




### 5.1 Input Field




Location:

Upper portion of the application window.

Purpose:

Allows the user to enter numbers that will be sorted.

Instructions:

1. Click inside the input field.
2. Type numbers separated by spaces.
3. Ensure that only numerical values are entered.

Example:

170 45 75 90 802 24

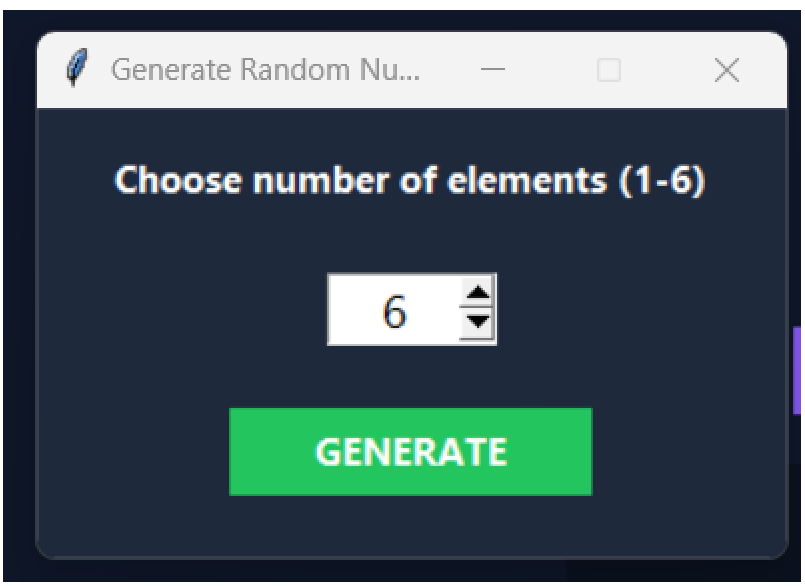
Valid Input:

- ✓ 10 20 30 40
- ✓ 5 1 8
- ✓ 100 25 75

Invalid Input:

- 10 abc 20
- 5,10,15
- 10@20

**5.2 RANDOM Button**

Purpose:

Generates random numbers automatically.

How to Use:

1. Click the RANDOM button.
2. A popup window will appear.
3. Select the desired number of elements.
4. Click GENERATE.

The application will automatically insert the generated numbers into the input field.

**5.3 START Button**


Purpose:

Begins the sorting visualization.

How to Use:

1. Enter numbers manually or generate random numbers.
2. Click START.

The application will:

- Validate the input
- Begin the Radix Sort process
- Generate sorting steps
- Display the visualization automatically
- Record activity logs

**5.4 STOP Button**


Purpose:

Pauses the ongoing animation.

How to Use:

1. While the visualization is running, click STOP.
2. The animation will pause immediately.
3. The button label changes to RESUME.

Use this feature when reviewing a particular sorting step.

**5.5 RESUME Button**


Purpose:

Continues a paused animation.

How to Use:

1. Click RESUME.
2. The visualization continues from the current step.

**5.6 PREVIOUS Button**


Purpose:

Displays the previous sorting step.

How to Use:

1. Click PREVIOUS.
2. The application moves back one step.
3. Previous array states become visible.

This feature allows detailed examination of earlier operations.

**5.7 NEXT Button**


Purpose:

Displays the next sorting step.

How to Use:

1. Click NEXT.
2. The application advances to the next recorded step.

This allows manual navigation through the sorting process.

**5.8 CLEAR Button**


Purpose:

Resets the application.

How to Use:

1. Click CLEAR.

The application will:

- Remove all entered values
- Clear the visualization area
- Clear output logs
- Reset all sorting steps

The system returns to its initial state.

# 6. VISUALIZATION AREA EXPLANATION

The visualization area displays the graphical representation of the sorting process.

The following components are shown.

**ORIGINAL ARRAY**


Purpose:

Displays the numbers currently being processed.

Description:

Each number appears inside a colored box.

Example:

170 45 75 90 802 24

The original array allows users to compare the current state of the data with the output array.

**OUTPUT ARRAY**


Purpose:

Displays the temporary arrangement of numbers during sorting.

Description:

As the algorithm processes digits, elements are placed into the output array.

The output array changes throughout the sorting process until the final sorted arrangement is achieved.

**COUNT ARRAY**


Purpose:

Displays the frequency and cumulative count values used by Counting Sort.

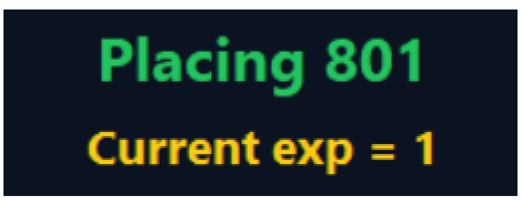
Description:

The count array contains ten positions representing digits 0 through 9.

Each box stores the count corresponding to a specific digit.

This helps users understand how positions are calculated during sorting.

**CURRENT EXPONENT VALUE**




Purpose:

Indicates which digit position is currently being processed.

Examples:

exp = 1 → Ones digit

exp = 10 → Tens digit

exp = 100 → Hundreds digit

The exponent changes automatically as the algorithm progresses.

# 7. OUTPUT LOG PANEL


The Output Log Panel displays detailed information regarding each sorting step.

Information displayed includes:

- Current Step Number
- Current Phase
- Output Array Status
- Start Messages
- Pause Messages
- Resume MessagesCompletion Messages

Example:

STEP 1

Phase: Counting Digits

Output: [170, 45, 75, 90, 802, 24]

This information helps users track the algorithm's progress.

# 8. COMPLETE USER WORKFLOW

Follow the procedure below to use the application successfully.

***Step 1***

Launch the application.

***Step 2***

Enter numbers manually or generate random values.

***Step 3***

Verify that the values are correct.

***Step 4***

Click START.

***Step 5***

Observe the automatic visualization.

***Step 6***

Review the Original Array.

***Step 7***

Review the Output Array.

***Step 8***

Observe updates in the Count Array.

***Step 9***

Use STOP if you wish to pause.

***Step 10***

Use RESUME to continue.

***Step 11***

Use PREVIOUS and NEXT to inspect specific steps.

***Step 12***

Wait until the completion message appears.

***Step 13***

Review the final sorted result.

***Step 14***

Click CLEAR if you want to perform another sorting operation.

# 9. ERROR MESSAGES

**Empty Input**

Condition:

User clicks START without entering any values.

Message:

“Enter numbers.”

Solution:

Enter one or more numerical values before clicking START.

**Invalid Input**

Condition:

Input contains letters or symbols.

Example:

10 20 abc

Message:

“Invalid input.”

Solution:

Enter numbers only.

**Maximum Elements Exceeded**

Condition:

More than six values are entered.

Message:

“Maximum of 6 elements only.”

Solution:

Reduce the number of entered values.

## 10. TIPS FOR USERS

• Enter numbers separated by spaces only.

• Do not use commas or special characters.

• Use the RANDOM button for quick demonstrations.

• Use STOP when analyzing a particular step.

• Use PREVIOUS and NEXT for detailed review.

• Use CLEAR before entering a new set of values.

• Observe the Count Array to understand how Radix Sort determines positions.

## 11. TROUBLESHOOTING

Problem: Application does not start.

Possible Cause:

Python is not installed.

Solution:

Install Python and verify installation.

Problem: Error message appears during execution.

Possible Cause:

Invalid input format.

Solution:

Enter numbers only.

Problem: Sorting does not begin.

Possible Cause:

Input field is empty.

Solution:

Enter values and click START again.

## 12. CONCLUSION

The Radix Sort Visualizer is an educational application that enables users to observe and understand the Radix Sort algorithm through interactive visualization. By providing manual controls, graphical representations, and detailed activity logs, the system offers a user-friendly environment for studying the sorting process. Following the procedures outlined in this manual will allow users to operate the application effectively and fully utilize all of its features.